

BARISCAN KOSE

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Education	<i>IoT & Machine Intelligence Graduate Certificate Program</i> <i>Sheridan College, ON, Canada</i>	<i>[09/2023 - 04/2024]</i>
	Engineered a smart irrigation system with ESP-32 and moisture sensors; acquired proficiency in FANUC robotics and PLC programming; utilized RTOS concepts on MPLAB X IDE, ICD3 debugger and PIC24 motherboard.	
	<i>Bachelor of Science in Electronics Engineering</i> <i>Sabanci University, Istanbul, Turkey</i>	<i>[09/2017 – 06/2022]</i>
	- Machine Learning, Microprocessor-Based System Design, Biomedical Instrumentation, Signal Processing, Cryptography, Computer Vision, Semiconductor Physics, MEMS. - Placed top %0.1 in national university-placement examination leading to full merit-based scholarship. - GPA 3.23/4.00 (Dean’s List: High Honor)	
Skills	Programming: C/C++, Assembly, Verilog, MATLAB, Shell (Makefile), Python, PyTorch, TensorFlow, SQL, Java, Spring Boot, JavaScript, React, HTML/CSS. Frameworks & Tools: RTOS, Linux (Ubuntu, Kali), Docker, Git, Jira, TCP/IP protocols, Wi-Fi, I2C, SPI, UART, FPGA, ARM Cortex, PIC microcontrollers, ESP32, Arduino, oscilloscopes, multimeters, LTSpice, Simulink, Advanced Design System, Xilinx, IAR Embedded Workbench IDE, Note Management Systems. Languages: English (IELTS C2), Turkish (Native), French (TEF B2)	
Work Experience	<i>Full-Stack Software Developer</i> <i>Electric Mind, Toronto, Canada</i>	<i>[05/2024 – 01/2026]</i>
	- Developed backend microservices for replatforming legacy vehicle fleet management using Java, Spring Boot, GraphQL, AWS (CloudFormation, Lambda, CodePipeline, CodeDeploy, IAM, Secrets), EDB, JDBC, SQL, Redis, and e2e tests. - Built full-stack applications with Django, HTMX, Tailwind, and React; designed SQL Server databases for brokerage hierarchy in master data management, ensuring efficient data workflows. - Implemented Salesforce solutions using sfdx, Apex, Lightning Web Components, and JavaScript; integrated Twilio recordings with OpenAI models for automatic call summarization and dynamic CRM updates. - Led generative AI projects with Azure AI Foundry, Azure Data Factory, BlandAI, Plaid, Python (Flask), and React; developed AI voice agents for operational automation and single-handedly designed a generative AI transaction search system with ML model comparisons.	
	<i>Embedded Software Engineer</i> <i>Airties, Istanbul, Turkey</i>	<i>[03/2023 – 08/2023]</i>
	- Spearheaded the development and refinement of firmware for WiFi extenders, enhancing device connectivity and performance using C/C++, Linux, and Shell scripting. - Prepared and presented task progress updates in weekly status meetings to team members and global partner companies across Asia, Europe & North America. - Advanced AT&T All-Fi Hub device capabilities through the integration of cutting-edge networking protocols and algorithmic enhancements, leveraging GIT, SVN, Broadcom BSP, HALs and TCP/IP expertise. - Elevated system reliability by employing GDB for rigorous debugging processes, resolving backtrace complexities, and effectively managing dynamic shared libraries.	

Data Scientist**[02/2022 – 03/2023]***Turing, California, USA (Remote)*

- Revolutionized data query processes using Mode Analytics and SQL, facilitating critical internal metric reporting and optimizing email marketing strategies for cross-functional teams of 20+ members.
- Played a pivotal role in third-party API integration, enhancing service functionality by scripting in Python and employing robust tools like Poetry, Jenkins, and Docker.
- Oversaw and improved data infrastructure integrity through strategic pipeline migrations, routine maintenance, and augmentation tasks, coordinated via GitHub and Shell scripts.

Machine Learning & Embedded Software Engineering Intern**[06/2021 – 08/2021]***Ideaktif Consultancy & Software LLC, Izmir, Turkey*

- Innovated an embedded SOS feature for ID cards utilizing ESP-32 motherboards and iBeacon-type beacons, significantly bolstering hospital emergency responses.
- Devised a predictive sales model with TensorFlow, dissecting variables such as time, pricing, and market positioning to enable informed business strategies.

Voluntary Work Civic Involvement Project**[Winter 2019]**

- Orchestrated and executed a series of programs at Metin Sabancı Schools for Special Education, tailored to enrich the lives of individuals with cerebral palsy and diverse disabilities.

Selected Projects Remote Beehive Monitoring System (Graduation Project)**[09/2021 – 05/2022]**

- Engineered an innovative beehive health monitoring solution, deploying temperature and humidity data acquisition via SHT-15 and LM35 sensors to a global AWS server using a cost-efficient SIM808 GPRS/GSM module paired with an Arduino UNO, aimed at mitigating colony collapse disorder.
- Nominated for the prestigious TUBITAK 1512 BiGG Uludag University Entrepreneurship Support Program for its potential impact on environmental sustainability.

Ensemble ML Pipeline for Stock Price Prediction with Technical Analysis on VPS**[03/2022]**

- Developed a sophisticated 3-tier ensemble machine learning pipeline on my virtual private server, integrating CatBoost, LightGBM, and Deep Neural Networks, complemented by a Local & Parallel Folding strategy, to harness technical & fundamental analysis for the prediction of US stock market trends.
- Implemented a robust data pipeline featuring meticulous pre- and post-processing steps, utilizing PostgreSQL, Ubuntu Linux and 3rd party APIs for continuous real-time market data extraction.

E-Commerce Website & Mobile Application**[Spring 2021]**

- Led the front-end development of an e-commerce web and mobile platform as part of a six-member Scrum team, delivering an engaging user experience through meticulous HTML, JS, CSS, ReactJS, and Redux implementations.
- Took the initiative in crafting a responsive and intuitive interface, contributing to the platform's market competitiveness and user satisfaction.

References

- References available upon request.